

Girish Nalawade

Backend Software Engineer | AI Engineer | Distributed Systems & GenAI

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EXPERIENCE

Arizona State University | Software Engineer

May 2025 – May 2026 | Tempe, AZ

- Built an AI-assisted **Care Connect** platform enabling voice-based patient interactions and structured clinical data capture, **reduced manual model validation time by 30%** with automated self-evaluation harnesses.
- Reduced LLM hallucinations and improved answer relevance in course Q&A by engineering a **hybrid RAG pipeline** combining semantic vector search with keyword retrieval, achieving **95% relevance** on evaluated question set.
- Accelerated feature delivery by using **Claude Code, Cursor agents, and GPT** as pair programmers for rapid prototyping, debugging, and test generation.

Western Union | Associate Software Engineer

Aug 2023 – Aug 2024 | Pune, India

- Built a Karate API automation suite with dynamic response chaining across **50+ internal REST APIs**, enabling end-to-end payment workflow validation from a single test input.
- Optimized the core transaction engine with Spring Boot, Redis caching, and async queues, **cutting payment latency by 30% (350 ms → 250 ms)** under production load.
- Containerized REST microservices on **Docker cutting deployment time by 40%** and using Kubernetes for improving cluster recovery time by 25% via health checks and rolling updates.
- Co-designed **T-View**, a real-time monitoring system for **1M+ daily transactions**, reducing mean fault recovery time by 40%.

Western Union | Software Engineering Intern

Jan 2023 – Aug 2023 | Pune, India

- Migrated **30+ microservices from Java 8 to Java 17**, modernizing configurations, hardening security, and improving startup time.
- Designed a serverless logging pipeline with AWS Lambda, S3, and CloudWatch, **cutting archival costs by 35%** across 20+ services.
- Load-tested services with JMeter at **10K concurrent requests**, identifying bottlenecks and improving fault tolerance by 33%.

EDUCATION

Arizona State University — M.S. in Computer Science (Software Engineering)

Aug 2024 – May 2026 | Tempe, AZ

Savitribai Phule Pune University — B.E. in Computer Engineering

Aug 2019 – May 2023 | Pune, India

PROJECTS

TrustYourself — Privacy-Focused AI Router | Presidio · spaCy · Ollama · FastAPI · Merkle Audit Log

Mar 2025 – Apr 2025

- Designed a privacy-preserving LLM router that redacts PII and keeps sensitive queries on a local model, sending only sanitized content to the cloud; **~80% PII detection accuracy at 600 ms/request**.
- Generated cryptographically signed receipts and a Merkle-based audit log so users can verify exactly what data left the device.

Provenant AI — AI Governance Platform | Next.js · NestJS · FastAPI · PostgreSQL

- Built a central registry for AI agents, holding each agent's configuration, architecture, and risk classification, enabling vulnerabilities and critical updates to be identified centrally, across **downstream agents**.
- Enabled cross-team reuse by allowing teams to review and adapt existing agents architectures, **cutting new-agent setup time by 75%** and reducing duplicated implementation effort.

AgentForge — Multi-Agent Orchestration Platform | FastAPI · React · FAISS

- Built a multi agent platform that runs **6 concurrent AI agents** independently for research, code analysis, and data processing, with role-based permissions and isolated execution sandboxes.

DeployQ — Automated Deployment & Debugging Platform | Docker · AWS ECR · S3 · Kafka · ClickHouse

- Engineered a self-service deployment system (Git URL → queued build → S3-served artifact) on Docker with ECR, **reducing manual deployment steps by 70%**; streamed build logs via Kafka into ClickHouse for querying.

ACHIEVEMENTS

- Research Paper:** Published “KG-ITP: A Knowledge Graph for Intelligent Travel Planning,” at IEEE COMPSAC 2025.
- Hackathon Winner:** Provenant AI, an AI governance platform for enterprise risk classification, controls, and audit readiness.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Go, SQL, C/C++

Backend & APIs: Spring Boot, FastAPI, Node.js/Express, gRPC, REST APIs, WebSockets, Microservices, Sentence-Transformers

Distributed Systems & Data: Kafka, Spark, Redis, PostgreSQL, MongoDB, Cassandra, Elasticsearch, RAG, LangChain, LangGraph

Cloud & DevOps: AWS (ECS, S3, Lambda, IAM), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, FAISS, pgvector, MCP